Agnibh Dasgupta

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science *2021 – 2026*
Utah State University, M.S., Computer Science *2018 – 2020*

Skills

Research:	Representation Learning, Watermarking, Computer Vision, NLP, Multimodal Learning
Models:	Transformers, ViT, CLIP, LLMs, Diffusion Models
ML Stack:	Python, PyTorch, TensorFlow, Hugging Face Transformers, CUDA
Training:	LoRA, DPO, GRPO, RLHF, RLAIF, PyTorch DDP, DeepSpeed, FSDP, Slurm/HPC
Agentic AI:	RAG, LangGraph, Tool Use, Function Calling, vLLM, Quantization
Evaluation:	LLM-as-a-Judge, RAGAS, A/B Testing, Human-in-the-Loop

Awards

CVPR Compute Transparency Champion Award 2026
- Awarded for exemplary reporting of computational resources, methodology, and reproducibility details

Professional Experience

- Led papers and experiments, hiring, interviewing, and mentoring interns in weekly meetings
- Worked in interdisciplinary teams with education researchers at UNL and doctors at Stanford
- Served as a peer reviewer, presented at conferences, authored NSF and graduate grant proposals

Selected Research and Projects

Siosa's Library: Path of Exile Q&A Agent *2026*
- Built a **RAG** based tool using **agentic AI** workflows to help new players navigate Path of Exile's steep learning curve
- Designed a **multi-agent** workflow using **LangGraph** to coordinate and tool selection across multiple data sources
- Hosted a pretrained **LLM** on **cloud infrastructure**, optimizing inference with **vLLM** and **quantization**
- Aligned model behavior via **human-in-the-loop post-training** (**DPO**, **PPO**, **GRPO**), evaluated with **LLM-as-judge**

Invariant Features in LLMs: Geometric Characterization and Model Attribution *(under review)* *2026*
- Investigated whether **LLMs** encode causally meaningful **representations** that can be used for **model attribution**
- Identified **invariant subspaces** in 9 open-source **LLMs** (7-10B params) via contrastive sensitivity decomposition
- Validated semantic and nuisance components causally through next-token prediction interventions
- Achieved 90%+ **zero-shot model attribution** accuracy across base, **fine-tuned**, and **distilled LLMs**

TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation *CVPR 2026*
- Addressed the challenge of maintaining **watermark** robustness under real-world camera capture conditions
- Built a **zero-watermarking** framework combining **CLIP** image–text features with a learned **augmentation** module
- Trained a differentiable auto-augmentor to learn camera-like distortions, replacing hand-designed **augmentation**
- Achieved 95%+ watermark recovery on **ImageNet** and **Visual Genome** under real-world camera conditions

Watermarking Language Models through Language Models *IEEE TAI 2025*
- Addressed the need for **LLM watermarking** that generalizes across architectures without modifying model weights
- Built a model-agnostic, prompt-guided **watermarking** scheme for **LLM**-generated text
- Tested on 25 open-source **LLMs** including **fine-tuned** and **distilled** variants using **LoRA** and **DeepSpeed**
- Achieved 88%+ detection accuracy under paraphrasing, **fine-tuning**, and **adversarial prompt attacks**

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning *CSCI 2023*
- Addressed the challenge of building a **watermarking** system robust to real-world noise via a learned representation
- Built a **ViT**-based **cross-attention watermarking** system with a **self-supervised** triplet loss objective
- Trained the encoder to cluster augmented views while separating different images in latent space
- Achieved 90%+ **bit recovery rate** (**BRR**), comparable to prior state-of-the-art under compound noise

Perspective Transformation Layer *CSCI 2022*
- Addressed the limitation of fixed preprocessing pipelines for handling viewpoint variation in image recognition
- Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines
- Evaluated on **SVHN**, Imagenette, and **MNIST** under viewpoint variation with minimal extra compute
- Achieved 96%+ classification accuracy, outperforming prior state-of-the-art on perspective benchmarks